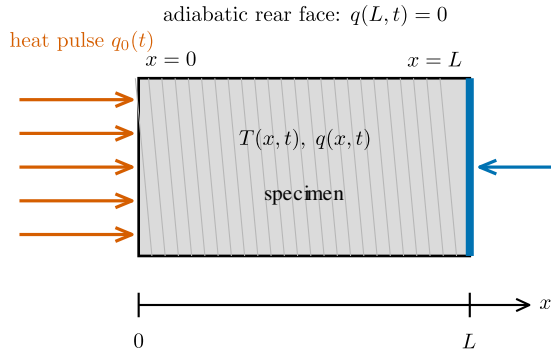


(a)



(b)

Guyer--Krumhansl system

$$\rho c \partial_t T = -\partial_x q$$

$$\tau_q \partial_t q + q = -\lambda \partial_x T + \kappa^2 \partial_{xx} q$$

Single equation (T-form)

$$\tau_q \partial_{tt} T + \partial_t T = \alpha \partial_{xx} T + \kappa^2 \partial_{xxt} T$$

Conditions

IC: $T(x, 0) = 0, q(x, 0) = 0$

BC: $q(0, t) = q_0(t), q(L, t) = 0$

pulse: $q_0 = 1 - \cos(2\pi t/t_p), 0 < t \leq t_p$

Resonance number

$$B = \kappa^2 / (\alpha \tau_q)$$

$B = 1$: Fourier solution recovered exactly